

1. Product Overview

One-line summary: A field-first web app that discovers real-world problems in a user’s chosen domain, ranks them by measurable pain, generates layered AI solutions, creates phased execution roadmaps with time estimates, and auto-produces developer-ready pre-build documents on demand.

Version: 1.0 (MVP)
Date: February 4, 2026
Owner: [Your Name/Team]
Status: Draft for MVP Development

2. Business Goals & Success Metrics

- **Primary Goal:** Enable founders, PMs, and engineers to go from domain selection to exportable roadmap + docs in <10 minutes, with 70% user satisfaction on artifact quality.
- **KPIs:**

Metric	Target (MVP Launch)	Measurement
Activation Rate	40% users create first roadmap in 7 days	Amplitude/PostHog
Time-to-Roadmap	<15 min median	User session tracking
Doc Quality Score	4.2/5 avg user rating	In-app NPS survey
Retention	30% D7, 15% D30	Cohort analysis
MRR (Post-MVP)	\$5K in 90 days	Stripe dashboard

3. Target Audience & User Personas

Persona	Needs	Pain Points
Founder/Solo	Validated ideas + solo roadmap	Idea overload, no execution plan
PM/Designer	Prioritized problems + UI specs	Vague stakeholder reqs
Engineer	Tech docs + API stubs	Missing architecture context
Domain Expert	Field-specific problems	Scattered forum noise

4. Core Features (MVP Scope)

Must-Have (Prioritized):

1. **Field-First Onboarding:** Multi-select domains (Education, Finance, etc.) → persists in profile.
2. **Problem Feed:** Ranked by Pain Index (Pi formula below), card UI with filters.
3. **Layered Solution Gen:** 6 layers (headline → tech stack → experiments) per problem.
4. **Roadmap Engine:** Phased output (6 default phases) with time estimates, export (MD/CSV/PDF).
5. **Doc Generators:** PRD + Tech Doc (one-click MD/PDF) with API stubs, schemas.
6. **Auth & Saves:** Clerk auth, saved problems/roadmaps list.

Out-of-Scope (Post-MVP): Payments, teams, community features, mobile apps.

5. Key Algorithms & Computations

Pain Index (Pi) Formula

$$Pi = \frac{Volume_{mentions} \times Sentiment_{complexity}}{Age_{hours}} \times Authority_{source}$$

- **Volume:** Unique mentions (7-day window).
- **Sentiment_Complexity:** (1-negativity) × semantic depth (0-1).
- **Age:** Hours since first mention (cap at 8760).
- **Authority:** Source weights (Reddit=0.8, HN=1.2).

Implementation: Postgres for clusters, Pinecone for embeddings, cron job for re-ranking.

6. User Stories & Flows

ID	As a...	I want...	So that...	Acceptance Criteria
US-01	New user	Select domains on signup	Results are tailored	5+ domains, persists to profile, affects feed
US-02	Explorer	Browse ranked problems	I find high-pain ones	20+ cards, Pi bar, sort/filter works
US-03	Solver	Generate solution layers	I get actionable plan	6 layers rendered, editable, export button

ID	As a...	I want...	So that...	Acceptance Criteria
US-04	Builder	Click "Create Roadmap"	I get phased plan	6 phases, time estimates, CSV export validates
US-05	Dev	Generate Tech Doc	I get build-ready specs	Includes DB schema, 3+ API stubs, SVG diagram

Primary Flow:

1. Onboard → Problem Feed → Select Problem → View Solution → Create Roadmap → Generate Docs → Export.

Edge Cases:

- No problems in domain → "Expand search?" prompt.
- LLM timeout → Cached template + edit mode.
- High-traffic source → Rate-limited scraper fallback.

7. Non-Functional Requirements

Category	Requirement
Performance	Feed loads <2s, doc gen <30s
Scalability	1K DAU, auto-scale Vercel/Supabase

Category	Requirement
Security	Clerk MFA, rate-limit APIs (/solve: 10/min), no exposed keys
Accessibility	WCAG 2.1 AA, keyboard nav
Tech Stack	Next.js/Tailwind (FE), Supabase/Prisma (BE/DB), Claude/GPT (AI)

8. Dependencies & Risks

Risk	Likelihood	Mitigation	Owner
Scraped data quality	High	Manual curator UI + source authority map	Eng
LLM hallucinations	Med	Template-first + confidence flags	AI
Cost overrun	Low	Token caching, model fallback (Claude → GPT)	Ops

9. Milestones & Timeline (Solo Dev Assumption)

Phase	Duration	Key Deliverables	Success Criteria
Sprint 0	2w	Auth + basic feed	US-01 complete

Phase	Duration	Key Deliverables	Success Criteria
Sprint 1-2	4w	Full discovery + solutions	50+ test problems, $P_i > 0.6$
Sprint 3	3w	Roadmap + exports	End-to-end flow
Launch	1w	Beta + metrics	100 signups, 30% activation

Total MVP Timeline: 10 weeks.

10. Open Questions & Next Steps

- Primary LLM: Claude 3.5 Sonnet vs. GPT-5o? (Benchmark extraction accuracy.)
- Test domains: Confirm top 3 (e.g., Education, SaaS, Health).
- Beta cohort: 50 founders via HN/Reddit?

Action Items:

1. Wireframe feed + roadmap modal (Figma, 2 days).
2. POC Pain Index pipeline (1 domain, 3 days).
3. [Your Name] approve by EOD Feb 6.